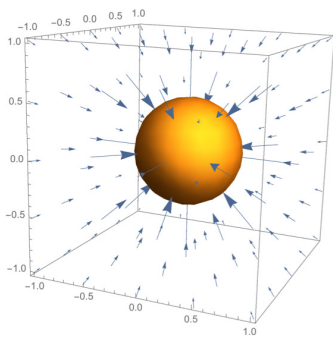


Lecture 16 Vector Fields

Def (Vector Fields) A vector field consists of a collection of vectors that are denoted by a vector-valued function $\vec{F}(x,y,z): \mathbb{R}^3 \rightarrow \mathbb{R}^3$, which assigns to each point (x,y,z) a vector labeled $\vec{F}(x,y,z)$.

$$\begin{aligned}\vec{r} &= \langle x, y, z \rangle & \vec{F}(x,y,z) &= \vec{F}(\vec{r}) = -\frac{GMM}{|\vec{r}|^3} \vec{r} = -GMM \frac{x\vec{i} + y\vec{j} + z\vec{k}}{(x^2 + y^2 + z^2)^{3/2}} \\ &= x\vec{i} + y\vec{j} + z\vec{k} & &= -GMM \left(\frac{x}{(x^2 + y^2 + z^2)^{3/2}} \vec{i} + \frac{y}{(x^2 + y^2 + z^2)^{3/2}} \vec{j} + \frac{z}{(x^2 + y^2 + z^2)^{3/2}} \vec{k} \right).\end{aligned}$$



$$\vec{F}(x,y,z) = F_x(x,y,z)\vec{i} + F_y(x,y,z)\vec{j} + F_z(x,y,z)\vec{k}.$$

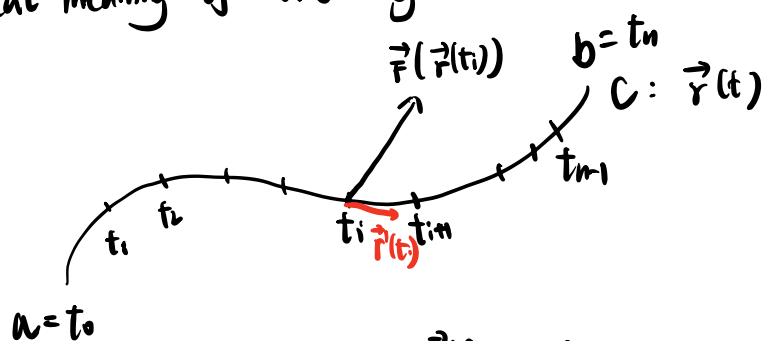
Line integrals of vector fields.

$$\begin{aligned}\text{Work} &= \text{force} \times \text{displacement} \\ &= \vec{F} \cdot \Delta \vec{r}\end{aligned}$$

Def. Given a continuous vector field $\vec{F}(x,y,z)$ and a path C which is parameterized by $\vec{r}(t)$, $a \leq t \leq b$, the line integral of $\vec{F}$ over C is defined to be $\int_a^b \vec{F} \cdot \vec{r}'(t) dt$.

$$\text{Notion: } \vec{r}'(t) dt = \frac{d\vec{r}}{dt} dt = d\vec{r} \quad \int_C \vec{F} \cdot d\vec{r}$$

Physical meaning of line integrals



$$\underbrace{\vec{F}(\vec{r}(t_i))}_{\text{force}} \cdot \underbrace{\vec{r}'(t_i) \Delta t_i}_{\substack{\text{velocity} \\ \text{time} \\ \text{displacement}}} \approx \text{work from } t_i \text{ to } t_{i+1}$$

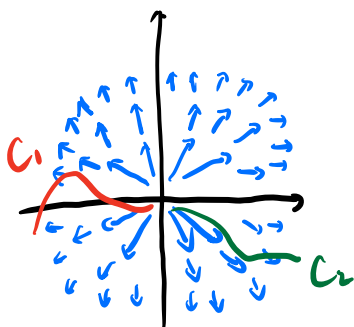
$$\int_a^b \vec{F} \cdot \vec{r}'(t) dt \approx \sum_i \vec{F}(\vec{r}(t_i)) \cdot \vec{r}'(t_i) \Delta t_i \quad \text{as } n \rightarrow \infty.$$

work done by the force $\vec{F}$ to move a particle along C

Geometric meaning of line integrals

$\int_C \vec{F} \cdot \vec{r}'(t) dt$ indicates whether the path C is overall flowing along or against the vector field.

$$\vec{F} \cdot \vec{r}'(t) = |\vec{F}| |\vec{r}'(t)| \cos \theta$$

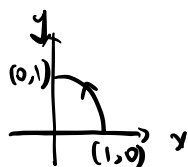


$$\int_{C_1} \vec{F} \cdot d\vec{r} < 0$$

$$\int_{C_2} \vec{F} \cdot d\vec{r} > 0$$

Circulation

Example. Let $\vec{F}(x,y) = -y\vec{i} + x\vec{j}$ and C be the counter-clockwise path along the unit circle from $(1,0)$ to $(0,1)$. Find $\int_C \vec{F} \cdot d\vec{r}$.



$$\text{Sol. } \vec{r}(t) = (\cos t)\vec{i} + (\sin t)\vec{j} \quad 0 \leq t \leq \frac{\pi}{2}$$

$$\vec{r}'(t) = (-\sin t)\vec{i} + (\cos t)\vec{j}$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^{\pi/2} \vec{F} \cdot \vec{r}'(t) dt$$

$$= \int_0^{\pi/2} (-\sin t \vec{i} + \cos t \vec{j}) \cdot (-\sin t \vec{i} + \cos t \vec{j}) dt$$

$$= \int_0^{\pi/2} (\sin^2 t + \cos^2 t) dt$$

$$= \frac{\pi}{2}$$

Example. Let $\vec{F}(x,y,z) = (y-x^2)\vec{i} + (z-y^2)\vec{j} + (x-z^2)\vec{k}$ and C be a path parametrized by $\vec{r}(t) = t\vec{i} + t^2\vec{j} + t^3\vec{k}$ where $0 \leq t \leq 1$.

Find $\int_C \vec{F} \cdot d\vec{r}$.

$$\text{Sol. } \int_C \vec{F} \cdot d\vec{r} = \int_0^1 \vec{F} \cdot \vec{r}'(t) dt$$

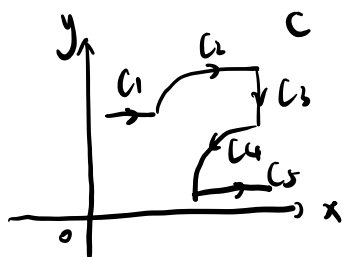
$$= \int_0^1 \langle y-x^2, z-y^2, x-z^2 \rangle \cdot \langle 1, 2t, 3t^2 \rangle dt$$

$$= \int_0^1 \langle 0, t^2 - t^4, t - t^6 \rangle \cdot \langle 1, 2t, 3t^2 \rangle dt$$

$$= \int_0^1 (2t^4 - 2t^5 + 3t^3 - 3t^8) dt$$

$$= \frac{2}{5} - \frac{2}{6} + \frac{3}{4} - \frac{3}{9}$$

$$= \frac{29}{60}.$$



$$\int_C \vec{F} \cdot d\vec{r} = \int_{C_1} \vec{F} \cdot d\vec{r} + \int_{C_2} \vec{F} \cdot d\vec{r} + \int_{C_3} \vec{F} \cdot d\vec{r} + \int_{C_4} \vec{F} \cdot d\vec{r} + \int_{C_5} \vec{F} \cdot d\vec{r}.$$

Independence of parametrization,

$$C: \vec{r}(z) \quad a \leq z \leq b$$

$$\vec{r}(t) \quad c \leq t \leq d$$

$$\vec{r}'(z) = \frac{d\vec{r}}{dz} = \frac{d\vec{r}}{dt} \frac{dt}{dz} = \vec{r}'(t) \frac{dt}{dz}$$

$$\begin{aligned} \int_{z=a}^{z=b} \vec{F} \cdot \vec{r}(z) dz &= \int_{z=a}^{z=b} \vec{F} \cdot \vec{r}'(t) \frac{dt}{dz} dz \\ &= \int_{t=c}^{t=d} \vec{F} \cdot \vec{r}'(t) dt. \end{aligned}$$

The work done by the force does not depend on how fast the particle travels.

Alternative notations for line integrals.

Arc-length parametrization $\vec{r}(s)$ s.t. $|\vec{r}'(s)| = 1$

$$\int_C \vec{F} \cdot \vec{r}'(s) ds \triangleq \int_C \vec{F} \cdot \vec{T} ds.$$

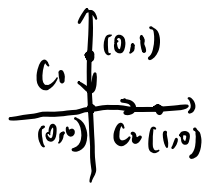
Differential form:

$$\vec{r} = x\vec{i} + y\vec{j} + z\vec{k} \quad d\vec{r} = dx\vec{i} + dy\vec{j} + dz\vec{k}$$

$$\vec{F} = F_x\vec{i} + F_y\vec{j} + F_z\vec{k}$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_C F_x dx + F_y dy + F_z dz.$$

Example. Evaluate $\int_C -y dx + x dy$
 where C is the path from $(0,1)$ down to $(0,0)$ along the y -axis, then to $(1,0)$ along the x -axis.



$$\text{sol. } \int_C -y dx + x dy$$

$$= \int_{C_1} -y dx + x dy + \int_{C_2} -y dx + x dy$$

Along C_1 , we have $x=0$, $dx=0$

$$\int_{C_1} -y dx + x dx = 0$$

Along C_2 , we have $y=0$, $dy=0$

$$\int_{C_2} -y dx + x dy = 0$$

$$\int_C -y dx + x dy = 0$$

$$\vec{F} = F_x \vec{i} + F_y \vec{j} + F_z \vec{k}$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot \vec{r}'(t) dt = \int_C \vec{F} \cdot \vec{T} ds = \int_C F_x dx + F_y dy + F_z dz.$$